Unit 2 Poster — Claim, Proof, Repair

Period B · Tue 11/10 + Thu 11/12 build, Fri 11/13 gallery walk · CED topics 2.1 to 2.9 (nothing binomial) · one DOK-3 claim per group

Printables (PDF): group sheets (six cards, two pages each; print one per group) · example board (project this first) · gallery-walk sheet (one per student, Friday) · teacher key (this guide, the board, the cards with answers, exit tickets, run of show, observer card, question and hint cards, rubric and scoring grid, answer key).

The hook for students: "Every claim on these cards is one somebody could really make about data we have already used. Three of them come straight from questions this class asked in the last two weeks. Your group's job is a verdict, the proof, and a repair. Then your poster teaches the room."

The six cards

#	Topics	The claim being judged	The mistake it busts (Period B evidence)	Data
1	2.1, 2.2	"Freeze Time people want to be rich: 26 chose Rich, more than any other power."	"The bigger count wins" **PERSISTENT B**: Period B's Unit 1 error, and 3 students read a relative-frequency bar chart as counts on quiz 2.1 (10/29).	Super Status table, 433 students
2	2.6	"Most students who want to be Famous chose Fly, so most students who chose Fly want to be Famous."	P(A | B) read as P(B | A). Asked in class 10/30: "why do the two conditional probabilities differ?"	Super Status table
3	2.5, 2.7	"Nobody who chose Telepathy wants to be Famous (0 of 68), so Telepathy and Famous are independent."	Mutually exclusive confused with independent. Asked 10/29; you promised the class an example of mutually exclusive events that are not independent. **PERSISTENT B**	Super Status table + two 2x2 tables the group designs
4	2.4, 2.7	"P(college grad or internet news) = 693/2500 + 687/2500 = 55.2%, and grads are no more likely than anyone else to get news online."	Addition rule without the subtraction; independence checked one way. 2.7 is B's least-finished lesson (worksheet mean 46; 8 of 14 not done).	2010 AP exam table, 2,500 adults (2.7 follow-along)
5	2.3, 2.8, 2.9	"A fair game! You win on 2 of every 6 rolls."	Expected value read as "every game"; winnings used instead of net; "fair" without asking for whom (2.8 exit-ticket traps). Also the 10/30 TI-84 promise.	Carnival die game (2.8 follow-along exit ticket)
6 spare	2.3	"6 heads in a row in 20 flips is impossible for a fair coin. It's rigged."	A simulation estimates; it never proves one outcome impossible (10/26: "was the simulated coin fair?").	randInt(0,1,20) on the TI-84

Coverage: 2.1 and 2.2 on card 1; 2.3 on cards 5 and 6; 2.4 everywhere (probabilities as relative frequencies, the complement region on card 4); 2.5 on card 3; 2.6 on cards 1, 2 and 4; 2.7 on cards 3 and 4; 2.8 and 2.9 on card 5. The walk sheet makes every student compute one number at every other poster, so everyone meets every topic.

Unit 2 Poster — groups, roles, days, materials

Groups

Class size	Groups	Who gets what
14 (Period B)	4 trios + 1 pair	The pair takes card 2 (two probabilities, the lightest build). A strong trio takes card 4 (the lesson most of the class has not finished) and another takes card 3 (the proof). Card 5 wants one calculator-confident student. Card 6 is the spare for a sixth group or early finishers.
12-13	4 groups	Drop card 1; its segmented bar chart becomes the Friday debrief demo.
15-18	6 groups	Card 6 goes to the sixth group.

Roles inside a trio

The whole group judges the claim together **first**, on scratch paper and whiteboards, before anyone touches the poster. Then:

- **Table Keeper** — every fraction and percent on the poster, with the arithmetic shown and the denominator named.
- **Visual** — the diagram the card asks for (segmented bar chart, tree, Venn, probability table and histogram, dotplot): title, labels, a key or scale.
- **Writer** — the verdict sentence, the repaired claim and the explanation sentences, in context, using the words from the language objective; checks the must-have list before the walk.

In a pair, the Table Keeper also draws the visual.

The three days (details on the run-of-show pages)

Tue 11/10 · 65 min · build	Thu 11/12 · 65 min · observed: defend, repair, finish	Fri 11/13 · 55 min · gallery walk
Do Now on boards. Objectives + EQ posted. Three Reads on the shared table. Example board. Cards out. Solve verdict + proof on scratch paper first, then start the poster. Exit ticket 1.	Do Now on boards. Defend round with question cards (groups 1-2, 3-4, 5 with you). Repair part of each card. Finish against the checklist. Five minutes on the 10/29 promise. Exit ticket 2, alone, on paper.	Final touches, posters up. Walk sheet at every other poster. Debrief on the two persistent ones. Score with the grid.

Materials and grading

Six poster boards (one spare), markers (four colors per group), rulers, tape, scratch paper, a whiteboard + marker + eraser for every student, TI-84s (at least two for card 5, ideally one per group), the printed cards (two pages, one per group), 14 walk sheets, 14 of each exit ticket, one question card per group, the hint cards, the board page (projected or printed large), the example board (projected). Each card's five must-haves are the rubric: all five = E, three or four = P, fewer = I. Scores E 115 / P 100 / I 90, any integer 90 to 115 per sheet; enter them with `node scripts/enter-poster.mjs` (format on the rubric page). Posters are 30% of Q2 Work for Period B.

AP Statistics · Period B · Unit 2 Poster

Tue 11/10 · Thu 11/12 · Fri 11/13 **DOK 3** CED 2026 topics 2.1-2.9 · skills 3.A, 3.B, 4.B

Learning objective

I can decide whether a claim about chance is supported by the data: find the probability the claim actually depends on (a conditional, a joint, a union, or an expected value), compare it to the claim, and justify my verdict in context.

Language objective

I can explain to my group and in writing on the poster why a probability claim is right or wrong, using the words **given, conditional, joint, union, mutually exclusive, independent** and **expected value** in complete sentences.

Essential question

When someone says "the odds are...", how do we know whether to believe them?

Frames

Verdict: "The claim says _____. The number it really depends on is _____ (a conditional P(___ | ___), a joint, a union, an expected value, or a simulated probability) = _____, not _____, so the claim is (supported / not supported / partly supported) because _____."
Partner talk: "I agree / disagree with _____ because _____." "Can you say that with the word *given*?"
Random variables: "In the long run, _____ per _____."

Word bank: given | conditional | joint | union | mutually exclusive | independent | expected value | long run | denominator | relative frequency | association

Today's job

- **Tuesday:** verdict and proof on scratch paper first. Then the poster.
- **Thursday:** defend your verdict to another group. Repair the claim. Finish against the checklist.
- **Friday:** gallery walk. One number at every poster.

Poster 1 — Who wants to be rich?

Topics 2.1-2.2 · Two-way tables, conditional relative frequencies, association

DOK 3

Your poster must show: the five "Rich" percents, a segmented bar chart (one 100% bar per superpower), your verdict, the repaired claim, and one association sentence.

Data (printed in the 2.4-2.6 follow-along)

Super Status! 433 high school students answered two questions: "What superpower would you choose?" and "What status would you prefer?"

Superpower	Famous	Happy	Healthy	Rich	Total
Fly	5	48	14	22	**89**
Freeze Time	3	63	9	26	**101**
Invisibility	3	62	11	22	**98**
Super Strength	4	43	17	13	**77**
Telepathy	0	49	0	19	**68**
Total	**15**	**265**	**51**	**102**	**433**

The claim you are judging

The student newsletter wrote: "Freeze Time people are the ones who want to be rich. 26 of them chose Rich, more than any other superpower."

DOK-3 — solve together before you build

a. For each superpower, find the percent of its fans who chose Rich. Write each as a fraction and a percent. Which total is the denominator each time, and why?

> **Answer.** Fly 22/89 = 24.7% · Freeze Time 26/101 = 25.7% · Invisibility 22/98 = 22.4% · Super Strength 13/77 = 16.9% · Telepathy 19/68 = 27.9%. The denominator is the row total: "given Fly" means only the 89 Fly students count. Overall, P(Rich) = 102/433 = 23.6%.

b. Draw a segmented bar chart: one bar per superpower, each bar 100% tall, split into Famous / Happy / Healthy / Rich. Add a key and a percent axis.

> **Answer** (percents within each superpower: Famous / Happy / Healthy / Rich). Fly 5.6 / 53.9 / 15.7 / 24.7 · Freeze Time 3.0 / 62.4 / 8.9 / 25.7 · Invisibility 3.1 / 63.3 / 11.2 / 22.4 · Super Strength 5.2 / 55.8 / 22.1 / 16.9 · Telepathy 0.0 / 72.1 / 0.0 / 27.9. Overall 3.5 / 61.2 / 11.8 / 23.6.

c. Verdict. Is the newsletter right about Freeze Time? Which superpower's fans are most likely to want to be rich, and by how many percentage points?

> **Answer.** The count is right (26 is the largest Rich count) but the reasoning is wrong: Freeze Time is the largest group (101 fans), so a large count is expected. By percent, Telepathy leads at 27.9%; Freeze Time is second at 25.7%, 2.2 points behind. Partly supported at best: wrong reason, wrong leader.

d. Repair. Rewrite the claim so it is true. Use a percent, not a count.

> **Answer** (any true version). "Among students who chose Telepathy, 27.9% want to be rich, the highest share of any superpower; Freeze Time fans are close at 25.7%."

e. Is there an association between superpower and status? Compare two bars that look different (try Telepathy and Fly) and explain in one sentence.

> **Answer.** Yes: the conditional distributions differ. Telepathy fans: 0% Famous, 0% Healthy, 72% Happy. Fly fans: 5.6% Famous, 15.7% Healthy, 54% Happy. With no association every bar would be split the same way.

Poster 1 — checklist, roles, sketch

Poster must-haves (= the rubric)

- ☐ Five "Rich" percents, each with its fraction and the row total as the denominator
- ☐ Segmented bar chart: five bars, each 100%, four segments, a key, a percent axis
- ☐ A verdict sentence that names the count AND the percent for Freeze Time and for the leader
- ☐ The repaired claim, true, with a percent
- ☐ One association sentence comparing two superpowers' bars

All five = E · three or four = P · fewer = I

The mistake this busts

In Unit 1 this class kept comparing counts from groups of different sizes: "the bigger count wins." Freeze Time has the most fans of any power (101), so of course it has a big Rich count. Percents *inside each row* are the fair comparison. On quiz 2.1, three of you also read a relative-frequency bar chart as if it showed counts.

Table Keeper
The five Rich fractions and percents, then all 20 percents for the bars. Show the arithmetic.

Visual
The segmented bar chart: five 100% bars, four segments, key, percent axis, title.

Writer
The verdict, the repaired claim, the association sentence. Use *given, conditional, association*.

Frame: "The claim compares _____ (counts). The probability it depends on is P(Rich | _____). Freeze Time: _____%. The leader is _____ at _____%. So the claim is _____ because _____."

given · conditional · relative frequency · association · denominator

Draft the poster sentences here. Every sentence names the context and uses a number from the table.

Poster 2 — The flipped headline

Topic 2.6 · Conditional probability: order matters

DOK 3

Your poster must show: both conditional probabilities with their denominators, a tree diagram or two shaded tables, the verdict on each half of the headline, the repaired headline, and a second flipped pair.

Data (printed in the 2.4-2.6 follow-along)

Super Status! 433 high school students answered: "What superpower would you choose?" and "What status would you prefer?"

Superpower	Famous	Happy	Healthy	Rich	Total
Fly	5	48	14	22	**89**
Freeze Time	3	63	9	26	**101**
Invisibility	3	62	11	22	**98**
Super Strength	4	43	17	13	**77**
Telepathy	0	49	0	19	**68**
Total	**15**	**265**	**51**	**102**	**433**

The claim you are judging

A headline in the school paper: "Most students who want to be Famous chose Fly. So most students who chose Fly want to be Famous."

DOK-3 — solve together before you build

a. Find P(Fly | Famous) and P(Famous | Fly). Write each as a fraction with the correct denominator, then as a percent. Which total is the denominator each time?

> **Answer.** P(Fly | Famous) = 5/15 = 33.3%: the 15 students who want to be Famous (a column total). P(Famous | Fly) = 5/89 = 5.6%: the 89 students who chose Fly (a row total). Same numerator, different "given" group.

b. Draw a tree diagram (first split Famous / not Famous, then Fly / not Fly) OR shade the "given" group on two copies of the table, to show why the denominators are 15 and 89.

> **Answer** (tree). Famous 15/433 = 3.5% → Fly 5/15; not Famous 418/433 → Fly 84/418 = 20.1%. Either visual earns the must-have if the two given groups are visibly different.

c. Verdict on each half. Is "most" right for the first sentence? (How big is 5 out of 15?) Does the second sentence follow from the first?

> **Answer.** First half: Fly is the top pick among the 15 Famous students (5; next is Super Strength with 4), but 33% is not "most": a plurality, not a majority. Second half: it does not follow; the condition is flipped. Only 5.6% of Fly-choosers want to be Famous.

d. Repair. Rewrite the headline as two true sentences, each with its probability.

> **Answer.** "Among the 15 students who want to be Famous, Fly was the most popular power (5, or 33%). Among the 89 who chose Fly, only 5.6% want to be Famous."

e. Build. Find another pair in the table where flipping the "given" changes the answer a lot. Write both probabilities and explain in one sentence why they differ.

> **Answer** (any correct pair). P(Happy | Super Strength) = 43/77 = 55.8% vs P(Super Strength | Happy) = 43/265 = 16.2%. P(Happy | Freeze Time) = 63/101 = 62.4% vs P(Freeze Time | Happy) = 63/265 = 23.8%. P(Rich | Telepathy) = 19/68 = 27.9% vs P(Telepathy | Rich) = 19/102 = 18.6%. Why: the numerator is the same cell; the denominator is the size of the given group.

Poster 2 — checklist, roles, sketch

The flipped headline

Poster must-haves (= the rubric)

- ☐ Both conditionals as fractions with the right denominators (15 and 89) and as percents
- ☐ A tree diagram or two shaded tables that show the two "given" groups
- ☐ A verdict on each half: "most" judged against 50%, and the flip named
- ☐ The repaired headline: two true sentences with their numbers
- ☐ A second flipped pair from the table with both probabilities and a one-sentence why

All five = E · three or four = P · fewer = I

The mistake this busts

On 10/30 someone asked why P(B | A) and P(A | B) come out different. The word after "given" names the group you are standing inside; that group's total is the denominator. Flip the given, flip the denominator, and the answer changes. Your teacher's chocolate example: P(17 | likes chocolate) = 20/23 but P(likes chocolate | 17) = 20/25.

Table Keeper
The two fractions and percents, and the second flipped pair. Name each denominator.

Visual
The tree (two levels, probabilities on every branch) or the two shaded tables.

Writer
The verdict on each half, the repaired headline, the why sentence. Use *given, conditional.*

Frame: "P(Fly | Famous) means: of the _____ students who want to be Famous, _____ chose Fly = _____%. P(Famous | Fly) means: of the _____ students who chose Fly, _____ want to be Famous = _____%. The headline flips _____, so _____."

given | conditional | denominator | plurality | majority

Draft the poster sentences here.

Poster 3 — Can't happen together, so unrelated?

Topics 2.5 + 2.7 · Mutually exclusive vs independent

DOK 3

Your poster must show: the joint probability, two independence tests, a verdict that uses both words, the general argument, and two 2x2 tables you designed.

Data (printed in the 2.4-2.6 follow-along)

Super Status! 433 high school students answered: "What superpower would you choose?" and "What status would you prefer?"

Superpower	Famous	Happy	Healthy	Rich	Total
Fly	5	48	14	22	**89**
Freeze Time	3	63	9	26	**101**
Invisibility	3	62	11	22	**98**
Super Strength	4	43	17	13	**77**
Telepathy	0	49	0	19	**68**
Total	**15**	**265**	**51**	**102**	**433**

The claim you are judging

"Nobody who chose Telepathy wants to be Famous: 0 of 68. Since the two never happen together, Telepathy and Famous are independent. One has nothing to do with the other."

DOK-3 — solve together before you build

a. Are "chose Telepathy" and "wants to be Famous" mutually exclusive? Show the joint probability P(Telepathy and Famous).

> **Answer.** P(Telepathy and Famous) = 0/433 = 0. They cannot both happen: mutually exclusive.

b. Are they independent? Test two ways: compare P(Famous) with P(Famous | Telepathy); compare P(Famous) × P(Telepathy) with P(Famous and Telepathy).

> **Answer.** P(Famous) = 15/433 = 3.5% but P(Famous | Telepathy) = 0/68 = 0%: not equal, so not independent. P(Famous) × P(Telepathy) = (15/433)(68/433) = 0.0054 but P(Famous and Telepathy) = 0: not equal, so not independent.

c. Verdict, in one sentence that uses both terms correctly. What does knowing "Telepathy" do to the chance of "Famous"?

> **Answer.** "Telepathy and Famous are mutually exclusive but not independent: knowing a student chose Telepathy drops the chance they want to be Famous from 3.5% to 0." Seeing one rules the other out, which changes its probability: that is dependence, not independence.

d. Prove it in general. Two events can each happen (each has probability above 0) and they are mutually exclusive. Explain why they can never be independent. (Hint: what does independence say P(A and B) must equal?)

> **Answer.** Independent means P(A and B) = P(A) × P(B). If both events are possible, that product is above 0. Mutually exclusive means P(A and B) = 0. Both cannot be true at once, so mutually exclusive events with positive probabilities are never independent.

e. Build. Make a 2x2 table of 100 students for "plays a sport" and "plays an instrument" where the two events are independent but NOT mutually exclusive (show the check). Then make a second table where they are mutually exclusive, run the independence check again, and say what you find.

> **Answer** (one of many). Independent: sport 60, instrument 50, both 30 → sport only 30, instrument only 20, neither 20. Check: P(instrument | sport) = 30/60 = 50% = P(instrument) = 50/100. Any table with both = P(S) × P(I) × 100 works. Mutually exclusive: sport 50, instrument 50, both 0, neither 0. Check: P(instrument | sport) = 0/50 = 0% but P(instrument) = 50%: not independent.

Poster 3 — checklist, roles, sketch

Can't happen together, so unrelated?

Poster must-haves (= the rubric)

- ☐ P(Telepathy and Famous) = 0 shown from the table, with the words *mutually exclusive*
- ☐ Independence tested two ways with the numbers (P(Famous) vs P(Famous | Telepathy); product vs joint)
- ☐ One verdict sentence that uses *mutually exclusive* AND *independent* correctly
- ☐ The general argument from part (d), written as a short proof that starts from what independence says P(A and B) must equal
- ☐ Two 2x2 tables (100 students each) with the independence check written under each

All five = E · three or four = P · fewer = I

The mistake this busts

On 10/29 the class asked for the difference between mutually exclusive and independent, and your teacher promised the class an example that settles it. Is "can't happen together" the same as "unrelated"? Your table decides, with numbers: test what knowing one event does to the probability of the other.

Table Keeper
The four probabilities, the product, and the numbers in both 2x2 tables.

Visual
The two 2x2 tables (must-have 5), each with its check written underneath; add a Venn diagram of the two events if there is room.

Writer
The verdict sentence and the general argument. Use *mutually exclusive, independent, joint*.

Frame: "P(Telepathy and Famous) = _____, so they are _____. But P(Famous) = _____ and P(Famous | Telepathy) = _____, which are (equal / not equal), so they (are / are not) _____. Knowing a student chose Telepathy _____ the chance they want to be Famous."

mutually exclusive · independent · joint · given · product

Draft the poster sentences here.

Poster 4 — Double counted

Topics 2.4 + 2.7 · Unions, complements, independence

DOK 3

Your poster must show: a Venn diagram (or two-way table) with all four regions, the corrected union, two independence tests, the sentence on why the two tests agree, and the "exact independence" number.

Data (printed in the 2.7 follow-along, from the 2010 AP exam)

A survey of 2,500 adults asked about education level and where they get their news.

Group	Count
College graduates	693
Get news mainly from the internet	687
College graduates who get news from the internet	245
All adults surveyed	**2,500**

The claim you are judging

A blog wrote: "P(college grad or internet news) = 693/2500 + 687/2500 = 55.2%. And college grads are no more likely than anyone else to get their news online."

DOK-3 — solve together before you build

a. Draw a Venn diagram (or a two-way table) and fill in all four regions: both, grad only, internet only, neither. Check that they add to 2,500.

> **Answer.** Both 245 · grad only 693 − 245 = 448 · internet only 687 − 245 = 442 · neither 2,500 − 1,135 = 1,365. Sum 2,500.

b. Find the mistake in the 55.2%. Compute the correct P(grad or internet). Which people were counted twice?

> **Answer.** The 245 adults who are both were counted in 693 and again in 687. P(grad or internet) = (693 + 687 − 245)/2500 = 1135/2500 = 45.4%. (The blog's 1380/2500 = 55.2% is a union with double counting.)

c. Test the second claim two ways: compare P(internet | grad) with P(internet); compare P(grad) × P(internet) with P(grad and internet). Verdict: are the events independent? Who is more likely to read news online?

> **Answer.** P(internet | grad) = 245/693 = 35.4% vs P(internet) = 687/2500 = 27.5%: not equal, not independent. P(grad) × P(internet) = 0.2772 × 0.2748 = 0.0762 vs P(both) = 245/2500 = 0.098: not equal, not independent. Grads are MORE likely to read news online (35.4% vs 27.5% overall, and vs 442/1807 = 24.5% of non-grads). The blog's second sentence is false.

d. Explain in one sentence why the two tests in (c) must always agree. (Hint: start from P(A and B) = P(A) × P(B) and divide both sides by P(A).)

> **Answer.** P(A and B) = P(A)P(B) is the same statement as P(A and B)/P(A) = P(B), which is P(B | A) = P(B): one equation rearranged, so one test fails exactly when the other does.

e. Build. Keep 693 college grads and 687 internet readers. What number in "both" would make the two events exactly independent? Is it possible with real people? What does that tell you?

> **Answer.** both = 693 × 687 / 2500 = 190.44 people: not a whole number, so no survey of these sizes is exactly independent; 190 or 191 would be as close as real people get. The observed 245 is far above 190.44, so the dependence is real, not rounding.

Poster 4 — checklist, roles, sketch

Double counted

Poster must-haves (= the rubric)

- ☐ Four regions filled and summing to 2,500 (both / grad only / internet only / neither)
- ☐ The group that was counted twice named, and the corrected union as a fraction of 2,500 and a percent
- ☐ Both independence tests with numbers, and a verdict naming who is more likely
- ☐ One sentence on why the two tests always agree
- ☐ The "both" count that would make the two events exactly independent, and what it tells you

All five = E · three or four = P · fewer = I

The mistake this busts

2.7 (Independent Events and Unions) is the lesson this class finished least. Its exit ticket set two traps, and both are hiding in this claim: one in the first sentence's arithmetic, one in how the second sentence was checked. Find both.

Table Keeper
The four regions, the union, both tests, the exact-independence count. Show every subtraction.

Visual
The Venn diagram with the overlap shaded (or the 2x2 table), every region labeled with its count.

Writer
The double-count sentence, the verdict, the why-the-tests-agree sentence. Use *union, joint, independent*.

Frame: "The 55.2% went wrong because _____. The correct P(grad or internet) = _____. P(internet | grad) = _____ and P(internet) = _____, so the events are _____, and _____ are more likely to read news online."

union · joint · independent · given · complement

Draft the poster sentences here.

Poster 5 — Is the game fair?

Topics 2.8, 2.9 + 2.3 · Random variables, mean and SD, simulation

DOK 3

Your poster must show: the probability distribution of net profit, the mean and standard deviation with work, both interpretations, your verdict, the fair fix, and your simulation average.

Data (the 2.8 follow-along exit ticket)

A carnival game costs $2 to play. You roll a fair six-sided die. Roll a 6: you win $10. Roll a 1: you win $3. Otherwise: you win nothing.

The claim you are judging

The sign on the booth: "A FAIR GAME! You win on 2 of every 6 rolls."

DOK-3 — solve together before you build

a. Let X = the player's net profit for one game (winnings minus the $2). List every value of X with its probability in a table. Check that the probabilities add to 1.

> **Answer.** X = +8 (roll 6) with 1/6 · +1 (roll 1) with 1/6 · −2 (roll 2, 3, 4, 5) with 4/6. Sum 6/6 = 1. Using $10, $3, $0 instead of net is the first trap.

b. Compute the mean μ_X and the standard deviation σ_X. Show the formula, the substitution, and the arithmetic.

> **Answer.** $\mu_X = 8(1/6) + 1(1/6) + (-2)(4/6) = (8 + 1 - 8)/6 = 1/6 \approx \$0.17$. $\sigma_X^2 = (8 - 1/6)^2(1/6) + (1 - 1/6)^2(1/6) + (-2 - 1/6)^2(4/6) = 485/36 \approx 13.47$, so $\sigma_X \approx \$3.67$. Shortcut: $E(X^2) = (64 + 1 + 16)/6 = 13.5$, minus $\mu^2 = 1/36$.

c. Interpret μ_X and σ_X in context. Use "in the long run" and say "per game".

> **Answer.** In the long run, over many games, a player gains about 17 cents per game on average. A single game's net typically differs from that average by about $3.67.

d. Verdict. Is the game fair? For whom? The sign's "2 of every 6 rolls" is really a 2/6 chance on each roll, not a promise: why is it still the wrong number to look at? Who does the sign fool, and who does the game actually favor?

> **Answer.** A fair game has $\mu_X = 0$. Here $\mu_X = +\$0.17$, so the game favors the PLAYER; the carnival loses about 17 cents per game on average. "2 of 6" is the chance of winning anything on one roll (not a guarantee of two wins in six), and it is not the expected net: it ignores the sizes of the payouts. The sign fools a player into thinking "only 2 of 6, the house must win", and it fools the carnival owner too.

e. Repair. Change ONE payout so that $\mu_X = 0$ exactly. Show the new table and the new mean.

> **Answer** (either). The 6 pays $9 (net +7): $\mu = (7 + 1 - 8)/6 = 0$. Or the 1 pays $2 (net 0): $\mu = (8 + 0 - 8)/6 = 0$. Raising the price to $13/6 \approx \$2.17$ also balances it (exactly only at $13/6) but is not a payout change.

f. Simulate. On the TI-84, randInt(1,6) is one game. Play 20 games, record the net each time, and find your average. Write it next to μ_X. Why do they differ, and what would happen with 1,000 games?

> **Answer.** Twenty-game averages vary a lot (their standard deviation is about $3.67/\sqrt{20} \approx \$0.82$), so most groups land somewhere between about $-\$1.50$ and $+\$1.80$. With 1,000 games the average settles near \$0.17: the law of large numbers.

Poster 5 — checklist, roles, sketch

Poster must-haves (= the rubric)

- ☐ Distribution table of X = net profit (not winnings) with probabilities that sum to 1, plus a probability histogram of X
- ☐ μ_X and σ_X with the formula, the substitution and the arithmetic shown
- ☐ Both interpretations in context, with "in the long run" and "per game"
- ☐ A verdict naming who the game favors and why "2 of 6" is the wrong number
- ☐ The one-payout fix with the new mean shown, and the dotplot of your 20 simulated games with its average marked next to μ_X

All five = E · three or four = P · fewer = I

The mistake this busts

Three traps from the 2.8 exit ticket: using the winnings instead of the NET (winnings minus the $2 you paid); reading the mean as what happens in every single game; and calling a game "fair" without asking fair to whom. And from 10/26: a simulation estimates; it does not give the exact answer.

Table Keeper
The table, μ, σ, the fix. Show the formula before the numbers.

Visual
A probability histogram of X, and a dotplot of your 20 simulated nets with their average marked.

Writer
The interpretations and the verdict. Use *expected value, long run*.

Frame: "X = _____ with probabilities _____. μ_X = _____, which means in the long run the player _____ about $_____ per game. σ_X = _____, so a single game typically ends about $_____ from the mean. The game favors the _____ because _____."

expected value · long run · net · fair · simulation

Draft the poster sentences here.

Poster 6 (spare) — Rigged coin?

Topic 2.3 · Estimating probabilities using simulation

DOK 3

Your poster must show: the simulation design, 25 recorded trials, a dotplot of the longest runs, the estimated probability, and your verdict.

Data

You make the data: 20 flips of a fair coin, simulated on the TI-84 with randInt(0,1,20), where 1 = heads. (MATH → PROB → randInt. ENTER repeats it.)

The claim you are judging

"*I flipped a coin 20 times and got 6 heads in a row. That can't happen with a fair coin. It's rigged.*"

DOK-3 — solve together before you build

a. Design. Write the simulation in three sentences: how one flip is modeled, what one trial is, and what you record from each trial (the longest run of heads in the 20 flips).

> **Answer.** One flip: a random digit 0 or 1, 1 = heads, each equally likely. One trial: 20 such digits, randInt(0,1,20). Record: the longest string of consecutive 1s in the trial.

b. Run 25 trials. Record the longest run each time. Draw a dotplot of the 25 values.

> **Answer.** Typical values 3 to 5; a 6 or more appears in roughly 3 of 25 trials.

c. Estimate P(longest run is 6 or more) from your dotplot.

> **Answer.** (number of trials with a run of 6 or more)/25. Expect something near 0.12; anything from 0.04 to 0.24 is normal variation for 25 trials.

d. Verdict. Is 6 in a row evidence of a rigged coin? Use your estimate.

> **Answer.** No. A run of 6 or more heads happens in about 12% of sets of 20 fair flips, roughly one set in eight: unusual, not impossible, not evidence of rigging.

e. Your estimate is not the exact answer. Why not? What would 1,000 trials do? (Your teacher has the exact value.)

> **Answer.** A simulation gives a relative frequency, which varies from run to run; more trials make it settle closer to the exact value 0.1223 (law of large numbers). For reference: a run of 5 or more, 0.250; 7 or more, 0.058; 8 or more, 0.027.

Poster 6 — checklist, roles, sketch

Poster must-haves (= the rubric)

- ☐ The design in three sentences (flip, trial, what is recorded)
- ☐ 25 trials recorded in a table
- ☐ A dotplot of the 25 longest runs with a scale and a title
- ☐ The estimated probability as a fraction of 25 and a decimal
- ☐ The verdict, plus one sentence on what more trials would do

All five = E · three or four = P · fewer = I

The mistake this busts

On 10/26 someone asked whether the simulated coin was fair. A simulation estimates a probability from many trials; it never proves a single outcome impossible. Rare is not impossible, and one surprising result is not proof of rigging.

Table Keeper
Runs the calculator, records the 25 longest runs, computes the fraction.

Visual
The dotplot with a scale that covers every value you got (expect 2 to 8; 0 and 20 are possible) and the "6 or more" region shaded.

Writer
The design sentences and the verdict. Use *simulation, relative frequency, long run*.

Frame: "One trial is _____. We recorded _____. In _____ of 25 trials the longest run was 6 or more, so we estimate P(run of 6 or more) ≈ _____. A run of 6 is _____, not _____, so the coin _____."

simulation · trial · relative frequency · long run · estimate

Draft the poster sentences here.

Exit ticket 1 — Tuesday (two per page, cut)

Exit ticket 1 · Tue 11/10 Name _____ Card #____

In your own words, no copying from the poster:
Our claim says _____.

_____

The number it really depends on is _____ (write it as P(...), P(... | ...), P(... or ...), μ, or estimated P(...)) = _____ (fraction, percent, or dollars).
So the claim is (supported / not supported / partly supported) because

_____

_____

_____

Use at least one of: given · joint · union · mutually exclusive · independent · expected value.

- cut here -

Exit ticket 1 · Tue 11/10 Name _____ Card #____

In your own words, no copying from the poster:
Our claim says _____.

_____

The number it really depends on is _____ (write it as P(...), P(... | ...), P(... or ...), μ, or estimated P(...)) = _____ (fraction, percent, or dollars).
So the claim is (supported / not supported / partly supported) because

_____

_____

_____

Use at least one of: given · joint · union · mutually exclusive · independent · expected value.

Exit ticket 2 — Thursday (two per page, cut)

Exit ticket 2 · Thu 11/12 Name _____ Alone, on paper.

1. In the Super Status table, nobody who chose Telepathy wants to be Healthy (0 of 68). A classmate says: "So Healthy and Telepathy are independent." True or false? One sentence why, with a number. (P(Healthy) = 51/433.)

2. Which is bigger, P(Happy | Super Strength) or P(Super Strength | Happy)? Compute both as fractions and percents. (Happy and Super Strength: 43. Super Strength total: 77. Happy total: 265.)

- cut here -

Exit ticket 2 · Thu 11/12 Name _____ Alone, on paper.

1. In the Super Status table, nobody who chose Telepathy wants to be Healthy (0 of 68). A classmate says: "So Healthy and Telepathy are independent." True or false? One sentence why, with a number. (P(Healthy) = 51/433.)

2. Which is bigger, P(Happy | Super Strength) or P(Super Strength | Happy)? Compute both as fractions and percents. (Happy and Super Strength: 43. Super Strength total: 77. Happy total: 265.)

Run of show

Tue 11/10 · 09:55-11:00 · build (65 min)

Min	Students	Teacher
0-5	Do Now on whiteboards: "From the table, P(Rich | Fly)?" Boards up on the cue.	Scan. Ask one board: "Why 89 and not 433?" (22/89 = 24.7%.)
5-13	Three Reads on the projected table. Read 1: what is the story (no numbers). Read 2: list the quantities on your board. Read 3: what claims could someone make from this table?	Post the objectives, the EQ, the frames and the word bank. Collect three claims from boards; do not judge them yet.
13-17	Watch the example board.	Talk-track (3 min): the word that names the group, the denominator, verdict, repair, seven parts.
17-20	Groups form; cards and materials out; roles chosen and written on the card.	Hand each group its card and a question card for Thursday.
20-38	Solve parts (a)-(c) on scratch paper and whiteboards. No poster yet.	Circulate with the why questions. At minute 28, tier-1 hint card to any blank board; tier-2 only after that.
38-60	Build: title, the claim, the data, the probability, the visual. Draft (d) and (e).	Check each group's verdict sentence before it goes on the board. Posters stay in the room.
60-65	Exit ticket 1, alone.	Collect. Sort by card for Thursday's defend round.

Thu 11/12 · 09:55-11:00 · observed: defend, repair, finish (65 min)

Min	Students	Teacher
0-6	Do Now on whiteboards: "True or false: P(Happy or Fly) = 265/433 + 89/433." Boards up.	Two why questions: "Who got counted twice?" "What is the correct number?" (False: (265 + 89 − 48)/433 = 306/433 = 70.7%.)
6-8	Read today's job on the board.	Objectives and language objective re-read aloud; today's goal: defend, repair, finish.
8-26	Defend round, partner groups 1-2, 3-4, 5 with the teacher. 4 min: group A presents its verdict with the frame; group B asks from the question card. 2 min: A revises its verdict sentence. Swap.	Time the rounds. Listen for the frame and the word bank. Ask one open why question per group.
26-50	Repair part of the card (d), (e) and card 5's (f) on the TI-84. Finish the poster against the must-have checklist; each must-have initialed by the role that owns it.	Circulate. Tier-2 hints only where a group is stuck on the repair. Check the checklist with each group.
50-56	Whole class at card 3's poster.	The 10/29 promise: Telepathy and Famous are mutually exclusive AND not independent, with the numbers (0 vs 3.5%). "Can't happen together rules the other one out, which changes its probability: dependence, not independence."
56-65	Exit ticket 2, alone, on paper.	Collect. These two items are the independent evidence for the day.

Run of show, Friday

Fri 11/13 · 08:55-09:50 · gallery walk (55 min)

Min	Students	Teacher
0-10	Final touches; posters up around the room.	Hand out walk sheets.
10-30	Walk: four rotations of 5 min, no docents; the poster must speak for itself (that is what the Writer's sentences are for). Start at the poster to your group's right. Work out the one number at every other poster on the walk sheet.	Signal rotations. Listen at the two persistent posters (2 and 3).
30-45	Debrief: hands for "which claim would have fooled you?"; two students read their sentence for posters 2 and 3.	Name the two rules: the given group is the denominator; mutually exclusive is not independent.
45-55	Hand in walk sheets; a fifth rotation for anyone who wants to revisit a poster.	Score with the grid (or after class); photos of the posters.

Observer card — Thursday 11/12

The Algebra 2 day-log format: title, body (Preparing / Interacting / Extending), frame, mode, minutes.

title	Unit 2 claim posters, day 2 of 3: defend, repair, finish (DOK 3)
body	**Preparing (8 min).** Do Now on whiteboards, a packet problem: true or false, P(Happy or Fly) = 265/433 + 89/433. Boards up. Two open questions ("Who got counted twice?" "What makes it right?"). Objectives and the language objective re-read. **Interacting (42 min).** Defend round in partner groups: four minutes to present the verdict with the frame while the listening group asks from the question card (why that denominator, what would make the claim true, say it with "given"), two minutes to revise the verdict sentence, swap. Then the repair part of each card: fix the headline, the one-payout fair game, the two 2x2 tables, the 190.4; card 5 simulates 20 games on the TI-84. Finish the poster against the five must-haves. Tiered hint cards go only to a stuck group. **Extending (15 min).** Whole class at poster 3: the promised example of events that are mutually exclusive but not independent, with the numbers. Exit ticket 2 alone on paper: a true-or-false on independence with a number, and two conditional probabilities compared.
frame	"The claim says ___. The number it really depends on is ___ (a conditional, a joint, a union, an expected value, or a simulated probability) = ___, not ___, so the claim is (supported / not supported / partly supported) because ___."
mode	groups of three (one pair); partner groups for the defend round; alone for the exit ticket
minutes	65

What the observer should count: open why questions asked; boards up at each cue; students who speak in the defend round; exit tickets collected.

Why questions (open; aim for more than half of all questions)

- Why is the denominator 89 and not 433?
- What would the claim need to say for your number to support it?
- Where did the 245 get counted twice?
- Why can't these two events be independent if they never happen together?
- What does +$0.17 promise a player, and what doesn't it promise?
- Which is the bigger group here, and why does that matter?
- If you only had the percents, could you still tell who is right?
- How would you convince someone who only looks at counts?
- What number in "both" would make these independent? Could that happen?
- Why does your simulation average differ from the mean?
- Can you say that again with the word "given"?
- Who does that sign fool, and who does it help?

Where each look-for lives on Thursday

Form item	Where
Objectives posted, content and language	The board page, up at 09:55; re-read at minute 6.
DOK level	DOK 3 on every card header and on the board: judge a claim, prove it, repair it.
Open questions, wait time	The why list; five seconds after each; boards up before anyone answers.
Language routine with frames	The verdict frame and the question cards in the defend round; word bank on the board.
Student-to-student discourse	Defend round, four minutes each way, every group speaks.
Whiteboards used	Do Now, repair scratch work, boards up at cues.
Visible independent work at the end	Exit ticket 2, alone, on paper, collected.
Scaffolds for multilingual learners	Word bank, sentence frames, tiered hint cards, the data table reprinted on every card, roles that give every student a job.
Evidence for feedback and adjustment	Exit tickets 1 and 2, the walk sheets and the scored posters, sorted by card.

Question cards (defend round) — one per group, cut

The listening group asks from its card. The presenting group answers with the frame. Nobody says "you're wrong"; you ask a question instead.

Question card

- Why that denominator?
- Which word in the claim did you test?
- What would make the claim true?
- Can you say it with the word "given"?
- Who does the number favor?
- Show me the one number that decides it.

Question card

- Why that denominator?
- Which word in the claim did you test?
- What would make the claim true?
- Can you say it with the word "given"?
- Who does the number favor?
- Show me the one number that decides it.

Question card

- Why that denominator?
- Which word in the claim did you test?
- What would make the claim true?
- Can you say it with the word "given"?
- Who does the number favor?
- Show me the one number that decides it.

Question card

- Why that denominator?
- Which word in the claim did you test?
- What would make the claim true?
- Can you say it with the word "given"?
- Who does the number favor?
- Show me the one number that decides it.

Question card

- Why that denominator?
- Which word in the claim did you test?
- What would make the claim true?
- Can you say it with the word "given"?
- Who does the number favor?
- Show me the one number that decides it.

Hint cards — tiered, hand out, never post

Tier 1 goes to a blank board at minute 28 on Tuesday. Tier 2 goes only to a group still stuck after tier 1, and only on the part they are stuck on. Frames come after thinking, not before.

Tier 1 (any card)

Find the word in the claim that names a group: *given, of, among, who, people*. Circle that group's TOTAL in the table. That total is your denominator. Now find the cell the claim is about and write the fraction.

Tier 2 · Card 1

Freeze Time row: 26 out of 101. Telepathy row: 19 out of 68. Which is bigger as a percent? Do all five rows the same way.

Tier 2 · Card 2

First sentence: you are inside the Famous COLUMN (15 people). Second sentence: inside the Fly ROW (89 people). Same cell, 5, on top of both.

Tier 2 · Card 3

Mutually exclusive asks: can both happen? (Look at the cell.) Independent asks: does knowing one CHANGE the chance of the other? Compare P(Famous) = 15/433 with P(Famous | Telepathy) = 0/68.

Tier 2 · Card 4

Draw two overlapping circles. Put 245 in the overlap FIRST. Grad-only = 693 − 245. Internet-only = 687 − 245. Neither = 2,500 minus everything inside the circles.

Tier 2 · Card 5

Net = winnings − 2. Rolls 2, 3, 4 and 5 all give the same net, so group them: probability 4/6. Mean = each net times its probability, added up.

Tier 2 · Card 6

MATH → PROB → randInt(0,1,20) ENTER gives 20 flips. Count the longest string of 1s. Press ENTER again for the next trial.

Rubric and scoring grid

Each card's five must-haves are the rubric. All five = E = 115. Three or four = P = 100. Fewer = I = 90. Any integer 90 to 115 is allowed per sheet (POSTER_WORK_TRACK_SPEC.md). Posters are 30% of Q2 Work for Period B; a missing poster is absent, never 0.

Card	M1	M2	M3	M4	M5	E / P / I	Score 90-115	Names
1 Who wants to be rich?								
2 The flipped headline								
3 Can't happen together								
4 Double counted								
5 Is the game fair?								
6 Rigged coin?								

Must-haves by card (M1 to M5)

Card	M1	M2	M3	M4	M5
1	Five Rich percents with fractions, row totals as denominators	Segmented bar chart: five 100% bars, four segments, key, percent axis	Verdict names count AND percent for Freeze Time and the leader	Repaired claim, true, with a percent	Association sentence comparing two bars
2	Both conditionals, denominators 15 and 89, percents	Tree or two shaded tables showing the given groups	Verdict on each half; "most" vs 50%; the flip named	Repaired headline, two true sentences with numbers	Second flipped pair with both numbers and a why
3	Joint probability 0, "mutually exclusive"	Two independence tests with numbers	Verdict using both terms correctly	General argument (ME + possible means not independent)	Two 2x2 tables with checks
4	Four regions summing to 2,500	Double count named; corrected union as fraction and percent	Both tests, verdict, who is more likely	Why the tests agree	Exact-independence count and what it means
5	Net distribution table + probability histogram, probabilities sum to 1	μ and σ with formula, substitution, arithmetic	Both interpretations, "long run", "per game"	Verdict: who it favors, why 2 of 6 is the wrong number	One-payout fix with $\mu = 0$, and the 20-game dotplot with its average beside μ
6	Design in three sentences	25 trials recorded	Dotplot with scale and title	Estimate as a fraction of 25 and a decimal	Verdict plus the more-trials sentence

Entering scores and grouping

Entering scores

File outside the repo (student names), then `node scripts/enter-poster.mjs <file>` to preview and `--apply` to write:
`unit=2 quarter=Q2 title=Unit 2 Poster (Claim, Proof, Repair) section=PeriodB`
`Real Name|115` or `Real Name|E` (E 115 / P 100 / I 90), one line per student.

Grouping grid (fill in before Tuesday)

Group	Card	Table Keeper	Visual	Writer
A				
B				
C				
D				
E (pair)	2		(Table Keeper)	

Answer key — Do Nows, exit tickets, the numbers, debrief notes

Do Nows (packet problems, whiteboards)

- **Tue:** P(Rich | Fly) = 22/89 = 24.7%. Denominator 89: inside the Fly row.
- **Thu:** False. P(Happy or Fly) = (265 + 89 − 48)/433 = 306/433 = 70.7%. The 48 students who are Happy AND Fly were counted twice in 354/433 = 81.8%.

Exit ticket 1 (Tue)

Full credit: the claim restated, the right number for that card (a conditional with the right denominator on cards 1-3, the corrected union and the conditional on card 4, the expected net in dollars on card 5, the estimated run probability on card 6), a verdict that matches the card's answer, one word from the bank used correctly. Sort by card; a group whose tickets disagree with each other gets the first visit Thursday.

Exit ticket 2 (Thu)

1. False. Healthy and Telepathy are mutually exclusive (0 of 68), not independent: P(Healthy) = 51/433 = 11.8% but P(Healthy | Telepathy) = 0/68 = 0%. Knowing Telepathy changes the chance.
2. P(Happy | Super Strength) = 43/77 = 55.8% is bigger than P(Super Strength | Happy) = 43/265 = 16.2%. Same cell (43), different given group (77 vs 265).

The numbers, card by card

Card	Key values
1	P(Rich | power): Fly 22/89 = 24.7%, Freeze Time 26/101 = 25.7%, Invisibility 22/98 = 22.4%, Super Strength 13/77 = 16.9%, Telepathy 19/68 = 27.9%; P(Rich) = 102/433 = 23.6%. Leader Telepathy by 2.2 points over Freeze Time. Segmented bars (Famous / Happy / Healthy / Rich, %): Fly 5.6 / 53.9 / 15.7 / 24.7 · Freeze Time 3.0 / 62.4 / 8.9 / 25.7 · Invisibility 3.1 / 63.3 / 11.2 / 22.4 · Super Strength 5.2 / 55.8 / 22.1 / 16.9 · Telepathy 0.0 / 72.1 / 0.0 / 27.9.
2	P(Fly | Famous) = 5/15 = 33.3% (plurality, not majority); P(Famous | Fly) = 5/89 = 5.6%. Spare pairs: 43/77 = 55.8% vs 43/265 = 16.2%; 63/101 = 62.4% vs 63/265 = 23.8%; 19/68 = 27.9% vs 19/102 = 18.6%.
3	P(Telepathy and Famous) = 0; P(Famous) = 15/433 = 3.5% vs P(Famous | Telepathy) = 0; P(F) × P(T) = 0.0054 vs 0. Independent 2x2: 60 sport, 50 instrument, 30 both (check 30/60 = 50/100). Mutually exclusive 2x2: 50 / 50 / 0 both (check 0/50 ≠ 50/100).
4	Regions 245 / 448 / 442 / 1,365. Union 1135/2500 = 45.4% (not 55.2%). P(internet | grad) = 245/693 = 35.4% vs P(internet) = 27.5% vs non-grads 442/1807 = 24.5%. Product 0.0762 vs joint 0.098. Exact independence needs 190.44 in both.
5	X: +8 (1/6), +1 (1/6), −2 (4/6). μ = 1/6 ≈ $0.17 for the player; σ^2 = 485/36 ≈ 13.47, σ ≈ $3.67. Fixes: 6 pays $9, or 1 pays $2. SD of a 20-game average ≈ $0.82.
6	P(run of 6+ heads in 20 fair flips) = 0.1223 (exact value, by computer); 5+: 0.250; 7+: 0.058; 8+: 0.027.

Debrief notes

- **The two persistent rules:** the word after "given" names the group whose total is the denominator (posters 1, 2, 4); mutually exclusive is the opposite of unrelated (poster 3, the 10/29 promise).
- **Card 1 nuance:** the newsletter's number is right and its reasoning is wrong; that is why the verdict frame has "partly supported".
- **Card 5 nuance:** fair means $\mu_X = 0$; "fair to whom" is the point. The 20-game averages will scatter; put them on one class dotplot and the law of large numbers teaches itself.
- **Worksheet note (not for students):** the 2.7 follow-along's union examples print Super Status numbers (265 + 29 − 15; 115/433 + 29/433) that do not match the printed table (Super Strength is 77, Freeze Time 101, Happy and Super Strength 43). Every number in this kit uses the printed table; the worksheet blanks need a fix in a later pass.